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Subject: Mathematics **CBSE CLASS 10 – WEEKLY CHAPTER TEST – 01** **Time:** 45 Minutes

Class: Class X (CBSE) **Chapter 4: Quadratic Equations** **MM: 20**

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, zero skipped steps, and final boxed answers with proper units.
2. In alternative (**OR**) sub-questions, complete step-by-step solutions and equal step marking apply to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Roots rejected by physical domain/restrictions are explicitly stated.

Official Evaluation Key | Maths Manthan Meerut

Section A: MCQ & Assertion-Reason Questions

MATHS MANTHAN | $5 \times 1 = 5$ Marks

Q1. Which of the following is a quadratic equation? [1]

(A) $x^2 + 2x + 1 = (4 - x)^2 + 3$

(B) $-2x^2 = (5 - x)\left(2x - \frac{2}{5}\right)$

(C) $(k + 1)x^2 + \frac{3}{2}x = 7$, where $k = -1$

(D) $x^3 - x^2 = (x - 1)^3$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(D)** $x^3 - x^2 = (x - 1)^3$

We know that a standard quadratic equation in one variable x has the form:

Formula Used: $ax^2 + bx + c = 0, \quad a \neq 0$

Checking each option by simplifying to standard polynomial form:

(A) $x^2 + 2x + 1 = (16 - 8x + x^2) + 3 \implies 10x - 18 = 0$ (degree 1, linear).

(B) $-2x^2 = 10x - 2 - 2x^2 + \frac{2}{5}x \implies \frac{52}{5}x - 2 = 0$ ($-2x^2$ cancels; degree 1, linear).

(C) Substituting $k = -1$: $(-1 + 1)x^2 + \frac{3}{2}x = 7 \implies \frac{3}{2}x - 7 = 0$ (coefficient of x^2 is 0).

(D) Expanding RHS using $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$:

$$x^3 - x^2 = x^3 - 3x^2 + 3x - 1 \implies 2x^2 - 3x + 1 = 0$$

Here $a = 2 \neq 0$ and highest power of x is 2.

Therefore, only **(D)** represents a quadratic equation. **[1 mark]**

Q2. Which of the following is **not** a quadratic equation? [1]

(A) $2(x-1)^2 = 4x^2 - 2x + 1$ **(B)** $2x - x^2 = x^2 + 5$

(C) $(\sqrt{2}x + \sqrt{3})^2 + x^2 = 3x^2 - 5x$ **(D)** $(x^2 + 2x)^2 = x^4 + 3 + 4x^3$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(C)** $(\sqrt{2}x + \sqrt{3})^2 + x^2 = 3x^2 - 5x$

Expanding and writing each option in the form $ax^2 + bx + c = 0$:

Formula Used: $ax^2 + bx + c = 0$ is NOT quadratic if $a = 0$ (or degree $\neq 2$)

(A) $2(x^2 - 2x + 1) = 4x^2 - 2x + 1 \implies 2x^2 + 2x - 1 = 0$ ($a = 2 \neq 0$, quadratic).

(B) $2x - x^2 = x^2 + 5 \implies 2x^2 - 2x + 5 = 0$ ($a = 2 \neq 0$, quadratic).

(C) Expanding LHS: $(2x^2 + 2\sqrt{6}x + 3) + x^2 = 3x^2 - 5x \implies 3x^2 + 2\sqrt{6}x + 3 = 3x^2 - 5x$.
Cancelling $3x^2$ from both sides gives $(2\sqrt{6} + 5)x + 3 = 0$, which is linear ($a = 0$).

(D) $x^4 + 4x^3 + 4x^2 = x^4 + 3 + 4x^3 \implies 4x^2 - 3 = 0$ ($a = 4 \neq 0$, quadratic).

Hence, option **(C)** is not a quadratic equation. **[1 mark]**

Q3. Which of the following equations has 2 as a root? [1]

(A) $x^2 - 4x + 5 = 0$ **(B)** $x^2 + 3x - 12 = 0$

(C) $2x^2 - 7x + 6 = 0$ **(D)** $3x^2 - 6x - 2 = 0$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(C)** $2x^2 - 7x + 6 = 0$

We know that $x = \alpha$ is a root of $p(x) = 0$ if and only if $p(\alpha) = 0$.

Formula Used: α is a root of $p(x) = 0 \iff p(\alpha) = 0$

Substituting $x = 2$ into each equation:

(A) $(2)^2 - 4(2) + 5 = 4 - 8 + 5 = 1 \neq 0$

(B) $(2)^2 + 3(2) - 12 = 4 + 6 - 12 = -2 \neq 0$

(C) $2(2)^2 - 7(2) + 6 = 8 - 14 + 6 = 0$

(D) $3(2)^2 - 6(2) - 2 = 12 - 12 - 2 = -2 \neq 0$

Since $x = 2$ satisfies $2x^2 - 7x + 6 = 0$, the correct choice is **(C)**. **[1 mark]**

Q4. If $\frac{1}{2}$ is a root of the equation $x^2 + kx - \frac{5}{4} = 0$, then the value of k is: [1]

(A) 2 **(B)** -2

(C) $\frac{1}{4}$ **(D)** $\frac{1}{2}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(A)** 2

Since $x = \frac{1}{2}$ is a root of $x^2 + kx - \frac{5}{4} = 0$, it must satisfy the equation:

Formula Used: If $x = \alpha$ is a root of $ax^2 + bx + c = 0$, then $a\alpha^2 + b\alpha + c = 0$

Substituting $x = \frac{1}{2}$:

$$\left(\frac{1}{2}\right)^2 + k\left(\frac{1}{2}\right) - \frac{5}{4} = 0 \implies \frac{1}{4} + \frac{k}{2} - \frac{5}{4} = 0 \implies \frac{k}{2} - 1 = 0 \implies \frac{k}{2} = 1 \implies \boxed{k = 2}$$

Therefore, option **(A)** is correct. **[1 mark]**

Q5. Assertion-Reason: [1]

Assertion (A): The quadratic equation $4x^2 - 12x + 9 = 0$ has two equal real roots.
Reason (R): A quadratic equation $ax^2 + bx + c = 0$ $(a \neq 0)$ has two equal real roots if its discriminant $D = b^2 - 4ac = 0$.

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true, but R is not the correct explanation of A.

(C) A is true, R is false.

(D) A is false, R is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).

Comparing $4x^2 - 12x + 9 = 0$ with $ax^2 + bx + c = 0$, we have $a = 4$, $b = -12$, and $c = 9$.

Formula Used: $D = b^2 - 4ac = 0 \implies$ two equal real roots

Computing the discriminant D:

$$D = (-12)^2 - 4(4)(9) = 144 - 144 = 0$$

Since $D = 0$, the equation $4x^2 - 12x + 9 = (2x - 3)^2 = 0$ has equal real roots $x = \frac{3}{2}, \frac{3}{2}$. Thus Assertion (A) is **true**.

Reason (R) gives the exact mathematical criterion $(D = 0)$ guaranteeing equal real roots, so Reason (R) is **true** and correctly explains Assertion (A). **[1 mark]**

Section B: Short Answer Questions — MATHS MANTHAN | 2 + 3 + 3 + 3 = 11 Marks

Q6. Find the roots of the quadratic equation $6x^2 - x - 2 = 0$ by the factorisation method. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

The given quadratic equation is $6x^2 - x - 2 = 0$.

Comparing with the standard form $ax^2 + bx + c = 0$, we have $a = 6$, $b = -1$, $c = -2$.

Formula Used: Split $bx = b_1x + b_2x$ such that $b_1 + b_2 = b$ and $b_1b_2 = ac$

Here, $ac = 6 \times (-2) = -12$ and $b = -1$. The required factors are -4 and $+3$ since $(-4) + 3 = -1$ and $(-4)(3) = -12$. **[$\frac{1}{2}$ mark]**

Splitting the middle term:

$$6x^2 - 4x + 3x - 2 = 0 \implies 2x(3x - 2) + 1(3x - 2) = 0 \implies (3x - 2)(2x + 1) = 0 \quad \textbf{[1 mark]}$$

Equating each linear factor to zero:

$$3x - 2 = 0 \implies x = \frac{2}{3} \qquad \text{or} \qquad 2x + 1 = 0 \implies x = -\frac{1}{2}$$

Hence, the roots of the equation are $\boxed{x = \frac{2}{3} \text{ and } x = -\frac{1}{2}}$. **[$\frac{1}{2}$ mark]**

Q7. Find two consecutive positive integers, sum of whose squares is 365. **[3]**

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Let the two consecutive positive integers be x and $x+1$, where $x > 0$ and $x \in \mathbb{Z}^+$.
According to the question, the sum of their squares is 365:

Formula Used: $(a+b)^2 = a^2 + 2ab + b^2$

$$x^2 + (x+1)^2 = 365 \implies x^2 + (x^2 + 2x + 1) = 365 \quad \textbf{[1 mark]}$$

Simplifying into standard quadratic form:

$$2x^2 + 2x - 364 = 0 \implies x^2 + x - 182 = 0 \quad [\tfrac{1}{2} \textbf{ mark}]$$

Using splitting the middle term ($14 \times (-13) = -182$ and $14 - 13 = 1$):

$$x^2 + 14x - 13x - 182 = 0 \implies x(x+14) - 13(x+14) = 0 \implies (x+14)(x-13) = 0 \quad \textbf{[1 mark]}$$

Therefore, $x = -14$ or $x = 13$.
Since the integers must be **positive** ($x > 0$), we reject $x = -14$.
Thus $x = 13$, and the next consecutive integer is $x+1 = 14$. (Verification: $13^2+14^2 = 169+196 = 365$.)
Hence, the required consecutive positive integers are $\boxed{13 \text{ and } 14}$. $[\tfrac{1}{2} \textbf{ mark}]$

Q8. Find the roots of the quadratic equation $4\sqrt{3}x^2 + 5x - 2\sqrt{3} = 0$ by factorisation. **[3]**

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given quadratic equation: $4\sqrt{3}x^2 + 5x - 2\sqrt{3} = 0$.
Comparing with $ax^2 + bx + c = 0$, we have $a = 4\sqrt{3}$, $b = 5$, and $c = -2\sqrt{3}$.

Formula Used: $ac = (4\sqrt{3})(-2\sqrt{3}) = -8 \times 3 = -24, \quad b = 5$

We look for two numbers whose sum is 5 and product is -24; these are $+8$ and -3 ($8 + (-3) = 5$, $8 \times (-3) = -24$). **[1 mark]**
Rewriting $3 = (\sqrt{3})^2$ and splitting the middle term:

$$\begin{aligned} 4\sqrt{3}x^2 + 8x - 3x - 2\sqrt{3} &= 0 \\ 4x(\sqrt{3}x + 2) - \sqrt{3}(\sqrt{3}x + 2) &= 0 \quad \textbf{[1 mark]} \\ (\sqrt{3}x + 2)(4x - \sqrt{3}) &= 0 \end{aligned}$$

Equating each factor to zero:

$$\sqrt{3}x + 2 = 0 \implies x = -\frac{2}{\sqrt{3}} = -\frac{2\sqrt{3}}{3} \qquad \text{or} \qquad 4x - \sqrt{3} = 0 \implies x = \frac{\sqrt{3}}{4}$$

Hence, the roots of the quadratic equation are $\boxed{x = -\dfrac{2\sqrt{3}}{3} \text{ and } x = \dfrac{\sqrt{3}}{4}}$. **[1 mark]**

Q9. A natural number, when increased by 12, equals 160 times its reciprocal. Find the number. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Let the required natural number be x ($x \in \mathbb{N}$, so $x > 0$). Its reciprocal is $\frac{1}{x}$.

Formula Used: $\text{Number} + 12 = 160 \times \frac{1}{\text{Number}}$

By the given condition:

$$x + 12 = \frac{160}{x} \quad \textbf{[1 mark]}$$

Multiplying both sides by x ($x \neq 0$):

$$x(x + 12) = 160 \implies x^2 + 12x - 160 = 0 \quad [\tfrac{1}{2} \textbf{ mark}]$$

Factorising by splitting the middle term ($20 \times (-8) = -160$ and $20 - 8 = 12$):

$$x^2 + 20x - 8x - 160 = 0 \implies x(x + 20) - 8(x + 20) = 0 \implies (x + 20)(x - 8) = 0 \quad \textbf{[1 mark]}$$

This gives $x = -20$ or $x = 8$.
Since x is a natural number ($x > 0$), we reject $x = -20$. (Check: $8 + 12 = 20 = 160 \times \frac{1}{8}$.)
Therefore, the required natural number is $\boxed{8}$. $[\tfrac{1}{2} \textbf{ mark}]$

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Q10. Case Study: Automotive Fuel Efficiency & Dual Vehicle Test Run

During an engineering endurance trial along a 400 km highway stretch, Johan and Jayant drive test vehicles at uniform speeds. Johan drives at a speed of x km/h, while Jayant drives 10 km/h faster than Johan. Jayant completes the 400 km journey in 2 hours less time than Johan. [4]

(i) Write expressions for the time taken by Johan and Jayant to cover 400 km in terms of x. **[1]**

(ii) Form the quadratic equation in x representing the above journey condition. **[1]**

(iii) Find the speeds of Johan and Jayant. **OR** Find the time taken by each driver to complete the 400 km run. **[2]**

► CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]

We know the standard distance–speed–time relation:

Formula Used: $\text{Time} = \dfrac{\text{Distance}}{\text{Speed}}$

(i) Time Expressions: Total distance = 400 km; Johan's speed = x km/h ($x > 0$) and Jayant's speed = $(x + 10)$ km/h.

$$\boxed{t_{\text{Johan}} = \frac{400}{x} \text{ hours}} \quad \text{and} \quad \boxed{t_{\text{Jayant}} = \frac{400}{x+10} \text{ hours}} \quad \textbf{[1 mark]}$$

(ii) Quadratic Equation: Since Jayant takes 2 hours less than Johan:

$$\frac{400}{x} - \frac{400}{x+10} = 2$$

$$400\left[\frac{(x+10)-x}{x(x+10)}\right] = 2$$

$$\frac{4000}{x^2+10x} = 2 \implies x^2 + 10x = 2000 \implies \boxed{x^2 + 10x - 2000 = 0} \quad \textbf{[1 mark]}$$

(iii) First Part (Speeds): Splitting the middle term ($50 \times (-40) = -2000$ and $50 - 40 = 10$):

$$x^2 + 50x - 40x - 2000 = 0 \implies (x+50)(x-40) = 0 \implies x = -50 \text{ or } x = 40$$

Since speed cannot be negative ($x > 0$), we reject $x = -50$. Thus $x = 40$ and $x + 10 = 50$.

Hence, $\boxed{\text{Johan's speed} = 40 \text{ km/h}}$ and $\boxed{\text{Jayant's speed} = 50 \text{ km/h}}$. **[2 marks]**

OR (Alternative Part — Times Taken): From part (iii) above, $x = 40$ km/h (after rejecting $x = -50$).

Substituting $x = 40$ into the time expressions obtained in part (i):

$$t_{\text{Johan}} = \frac{400}{40} = 10 \text{ hours}, \qquad t_{\text{Jayant}} = \frac{400}{40+10} = \frac{400}{50} = 8 \text{ hours}$$

Verification: $10 - 8 = 2$ hours.

Hence, $\boxed{\text{Johan takes 10 hours}}$ and $\boxed{\text{Jayant takes 8 hours}}$. **[2 marks]**

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